SIMATS ENGINEERING

MODEL PRACTICAL EXAMINATION – Aug._2026

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1709**

Register Number	**B.Bhavana**
Name	**192312428**

Set 1

1. Write a python program for 8-puzzle problem consists of a 3x3 grid containing 8 numbered tiles and one empty space. The objective is to move the tiles to achieve the goal state by sliding them into the empty space.
   Initial State:
   1 2 3
   4 6
   7 5 8
   Goal State:
   1 2 3
   4 5 6
   7 8
2. Write a python program for the following CRYPT arithmetic (or alphameric) puzzles involve replacing letters with digits to form valid arithmetic equations. Each letter represents a unique digit, and no number starts with a zero. For example:
   ```
     SEND
   + MORE
   ------------
    MONEY
   ```
   Where each letter represents a digit, and the sum must be valid. The same letter always represents the same digit.
3. Write a prolog program for monkey banana problem.
4. Write a Prolog program to find items and its location using following facts
   location(Record, Table).
   location(Spoon, kitchen).
   location(TV, Hall).

Laboratory Exam Component - 100 marks			
Programs (4)			
Description	**Marks Distribution**	**Total Marks**	**Marks Obtained**
Aim & Algorithm	2.5*4 = 10 Marks	10	
Programs	10*4 = 30 Marks	40	
Debugging	2.5*4 = 10 Marks	10	
Input & Output	05*4 = 20 Marks	20	
Result	2.5*4 = 10 Marks	10	
GitHub content	10 Marks	10	
Total Lab Exam Marks		**100**	

Set 2

1. Write a Python program to solve the following. You are given two jugs with capacities of 5 Liters and 4 Liters, respectively. Neither jug has any measurement markings. You need to measure exactly 3 Liters of water, using the following operations:

i) Fill either jug completely.
ii) Empty either jug completely.
iii) Pour water from one jug into the other until either the first jug is empty or the second jug is full.

2. Write a python program for the following Problem.

   Three missionaries and three cannibals need to cross a river using a boat that can hold at most two people. The following conditions must be satisfied: At no point can the number of cannibals on either side of the river outnumber the missionaries (if missionaries exist on that side). The boat cannot cross the river by itself; it needs at least one person to row it.

3. Write a Prolog program find a person is Mortal or NOT with the following facts. Socrates, Einstein, Alexander.

   man(socrates), man(Einstein), man(Alexander).

4. Write a Prolog program to find the number of vowels in the following fact. "I am an brilliant student in Saveetha University"

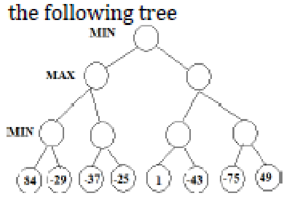
Laboratory Exam Component - 100 marks			
Programs (4)			
Description	**Marks Distribution**	**Total Marks**	**Marks Obtained**
Aim & Algorithm	2.5*4 = 10 Marks	10	
Programs	10*4 = 30 Marks	40	
Debugging	2.5*4 = 10 Marks	10	
Input & Output	05*4 = 20 Marks	20	
Result	2.5*4 = 10 Marks	10	
GitHub content	10 Marks	10	
Total Lab Exam Marks		**100**	

SET:3

Write a prolog program for the following problem “A monkey is in a room with a banana hanging from the ceiling. The monkey cannot reach the banana directly. However, there is a box in the room that the monkey can use to reach the banana. The problem involves a sequence of actions to enable the monkey to get the banana.
Initial State:
The monkey is on the ground.
The banana is hanging from the ceiling.
The box is in the room at a fixed location.
Goal State:
The monkey has the banana.
Allowed Actions:
Walk(Location): The monkey can walk to a new location.
Push(Box, NewLocation): The monkey can push the box to a new location if it is at the same location as the box.
Climb(Box): The monkey can climb on the box if they are at the same location.
Grasp(Banana): The monkey can grasp the banana if it is on the box and the banana is directly above.

2. Write a python program for Map colouring Problem.

3. Write a python program for Alpha beta Pruning to findthe optimal path to win the game for the following tree



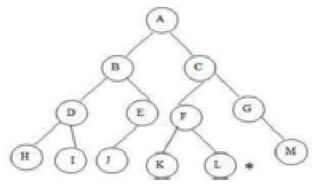
4.Write a prolog program for forward chaining for the following problem
It is illegal for a person to steal property belonging to others. Alice stole a painting from Bob, and it is known that stealing someone's property is a crime. Prove that "Alice is a criminal."

Laboratory Exam Component - 100 marks			
Programs (4)			
Description	**Marks Distribution**	**Total Marks**	**Marks Obtained**
Aim & Algorithm	2.5*4 = 10 Marks	10	
Programs	10*4 = 30 Marks	40	
Debugging	2.5*4 = 10 Marks	10	
Input & Output	05*4 = 20 Marks	20	
Result	2.5*4 = 10 Marks	10	
GitHub content	10 Marks	10	
Total Lab Exam Marks		**100**	

Set 4

1. Write a Python program to solve the TSP using the brute force approach by:
   Generating all possible permutations of the cities.
   Calculating the total distance for each permutation.
   Identifying the permutation with the shortest distance.

2. Write a Python program to find the goal L using Depth First Search using the following tree.



3. Write a Prolog program to print particular Bird can swim or not using following facts.
Bird (Eagle).
Bird (sparrow).
Bird (penguin).
Bird (duck
Swim (penguin) :- !, fail.
Fly(X) :- bird(X).

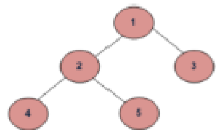
4. Write a Prolog program that predicts the type of an animal based on its characteristics. Use the predicates and the characteristics.

   If an animal has feathers and lays eggs, it is a bird.
   If an animal has fur and gives birth to live young, it is a mammal.
   If an animal has scales and lays eggs, it is a reptile.

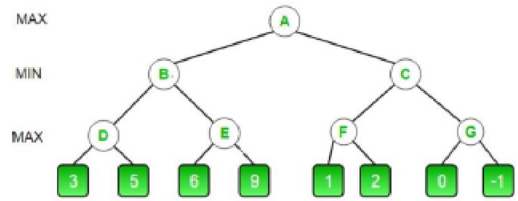
Laboratory Exam Component - 100 marks			
Programs (4)			
Description	**Marks Distribution**	**Total Marks**	**Marks Obtained**
Aim & Algorithm	2.5*4 = 10 Marks	10	
Programs	10*4 = 30 Marks	40	
Debugging	2.5*4 = 10 Marks	10	
Input & Output	05*4 = 20 Marks	20	
Result	2.5*4 = 10 Marks	10	
GitHub content	10 Marks	10	
Total Lab Exam Marks		**100**	

SET:5

1. Write a Python program for Limited Depth First search to reach the goal "3" with the limit of 1



2. Write a python program to find optimal maximum value from the following tree using alpha beta pruning.



3. Write a prolog program to Modify the Towers of Hanoi problem to include the following constraints:

   Time Restriction: Each disk move takes 1 unit of time. Calculate the total time required for NNN disks.

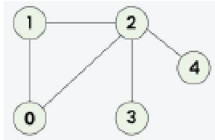
   Move Limit: Limit the number of allowed moves. If the task cannot be completed within the move limit, output "Move limit exceeded."

4. Write a prolog program for family tree

Laboratory Exam Component - 100 marks			
Programs (4)			
Description	**Marks Distribution**	**Total Marks**	**Marks Obtained**
Aim & Algorithm	2.5*4 = 10 Marks	10	
Programs	10*4 = 30 Marks	40	
Debugging	2.5*4 = 10 Marks	10	
Input & Output	05*4 = 20 Marks	20	
Result	2.5*4 = 10 Marks	10	
GitHub content	10 Marks	10	
Total Lab Exam Marks		**100**	

SET:6

1. Write a python program for depth first search for the following graph from the initial state 1 to goal state 4



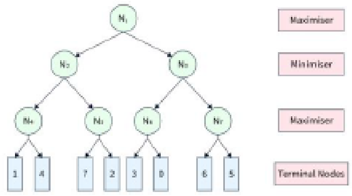
2. Write a python program to evaluate 4 queens problem.

3. Write a Prolog program to determine if a person should carry an umbrella based on the weather conditions. Use the following facts and rules:
   Facts:
   rainy(mumbai).
   rainy(delhi).
   windy(chennai).
   cold(delhi).
   Rules:
   If it is rainy or windy, a person should carry an umbrella.
   If it is cold and rainy, a person should wear warm clothes in addition to carrying an umbrella.
4. Write a Prolog program to predict the type of vehicle based on its properties using backtracking.
   Facts:
   type(car, four_wheeler).
   type(bike, two_wheeler).
   type(bus, four_wheeler).
   type(scooter, two_wheeler).
   fuel(car, petrol).
   fuel(bike, petrol).
   fuel(bus, diesel).
   fuel(scooter, petrol).
   Rules:
   A vehicle is identified based on its type (two_wheeler or four_wheeler) and its fuel type.
   The program should allow querying the vehicle type and fuel.

Laboratory Exam Component - 100 marks			
Programs (4)			
Description	Marks Distribution	Total Marks	Marks Obtained
Aim & Algorithm	2.5*4 = 10 Marks	10	
Programs	10*4 = 30 Marks	40	
Debugging	2.5*4 = 10 Marks	10	
Input & Output	05*4 = 20 Marks	20	
Result	2.5*4 = 10 Marks	10	
GitHub content	10 Marks	10	
Total Lab Exam Marks		100	

Set 7

1. Write a python program to find the pruned nodes using alpha beta pruning for the following tree



2. Design a web page of your own personnel details to show how Anchor tags and titles.
3. Write a Prolog program to implement book and author matching using the following facts:

Facts:
book_author(the_hobbit, tolkien).
book_author(the_fellowship_of_the_ring, tolkien).
book_author(harry_potter, rowling).
book_author(chamber_of_secrets, rowling).
book_genre(the_hobbit, fantasy).
book_genre(the_fellowship_of_the_ring, fantasy).
book_genre(harry_potter, fantasy).
book_genre(chamber_of_secrets, fantasy).
Task:
Write queries to:
Find all books written by a specific author (e.g., tolkien).
Find the genre of a specific book (e.g., the_hobbit).

4.Write a Prolog program to define relationships in a royal family tree with the following facts:

Facts:
king(george_v).
queen(elizabeth).
prince(charles).
prince(william).
princess(diana).
parent(george_v, elizabeth).
parent(elizabeth, charles).
parent(charles, william).
spouse(charles, diana).
spouse(william, kate).

1. Write rules to define grandparent, child, and sibling relationships.
2. Query the family tree to:
   - Find the grandparents of william.

Laboratory Exam Component - 100 marks			
Programs (4)			
Description	Marks Distribution	Total Marks	Marks Obtained
Aim & Algorithm	2.5*4 = 10 Marks	10	
Programs	10*4 = 30 Marks	40	
Debugging	2.5*4 = 10 Marks	10	
Input & Output	05*4 = 20 Marks	20	
Result	2.5*4 = 10 Marks	10	
GitHub content	10 Marks	10	
Total Lab Exam Marks		100	

Set 8

1. Write a Python program to solve the Map Colouring problem using Constraint Satisfaction Problem (CSP) for the following regions:

- Regions: {A, B, C, D, E}
- Adjacency constraints:
  - A is adjacent to B and C.
  - B is adjacent to A, C, and D.
  - C is adjacent to A, B, and E.
  - D is adjacent to B and E.
  - E is adjacent to C and D.
- Colours: {Red, Green, Blue}

Display a valid colouring of the regions such that no two adjacent regions share the same colour.

2. Write a Python program to apply the A* algorithm to solve the following tree problem:

- Nodes: {A, B, C, D, E, F}
- Edges and costs:
  - A → B (1), A → C (4)
  - B → D (2), B → E (5)
  - C → E (1), C → F (3)
- Heuristics h(n)h(n)h(n):
  - h(A) = 7, h(B) = 6, h(C) = 4, h(D) = 1, h(E) = 2, h(F) = 0

Find the optimal path from A (start) to F (goal).

3. Write a Prolog program to implement an Inference Engine that evaluates rules and facts based on logical conditions. Use the following facts and rules:

Facts:

```
rule(likes_milk, (mammal(X), not(carnivore(X)))).
rule(can_fly, (bird(X), not(penguin(X)))).
mammal(cow).
mammal(elephant).
carnivore(tiger).
bird(eagle).
bird(penguin).
```

Rules:

1. A mammal likes milk unless it is a carnivore.
2. A bird can fly unless it is a penguin.

Queries:

1. Determine if a cow likes milk.
2. Check if an eagle can fly

4.Write a Prolog program to diagnose faults in a car based on observed symptoms.

Predicates:

- diagnosis(Car, Fault): Predicts the fault in a car.
- problem(Car, Symptom): Observes the car's symptoms.

Facts:

```
problem(car1, engine_noise).
problem(car1, oil_leak).
problem(car2, battery_low).
problem(car2, slow_start).
problem(car3, flat_tire).
```

Rules:

```
diagnosis(car1, engine_failure) :- problem(car1, engine_noise), problem(car1, oil_leak).
diagnosis(car2, battery_issue) :- problem(car2, battery_low), problem(car2, slow_start).
diagnosis(car3, tire_puncture) :- problem(car3, flat_tire).
```

Laboratory Exam Component - 100 marks			
Programs (4)			
Description	Marks Distribution	Total Marks	Marks Obtained
Aim & Algorithm	2.5*4 = 10 Marks	10	
Programs	10*4 = 30 Marks	40	
Debugging	2.5*4 = 10 Marks	10	
Input & Output	05*4 = 20 Marks	20	
Result	2.5*4 = 10 Marks	10	
GitHub content	10 Marks	10	
Total Lab Exam Marks		100	

Set 9.

1. Write a Python program to solve a Sudoku puzzle using backtracking. Given a 9x9 grid with some cells pre-filled, your program should find a solution that satisfies the following constraints:
   Each row must contain the numbers 1–9 with no duplicates.
   Each column must contain the numbers 1–9 with no duplicates.
   Each of the 3x3 sub-grids must contain the numbers 1–9 with no duplicates.

Input:
An incomplete 9x9 Sudoku grid with 0 representing empty cells.
Output:
A completed grid satisfying all constraints.

2. Write a Python program to solve the N-Queens problem using backtracking.

- The objective is to place N queens on an N×N times N×N chessboard such that no two queens threaten each other.
- Queens cannot be in the same row, column, or diagonal.

3. Write a Prolog program to implement a knowledge base for animal classification based on characteristics.
Fact
has_feathers(eagle).
has_feathers(penguin).
lays_eggs(eagle).
lays_eggs(penguin).
lays_eggs(crocodile).
has_scales(crocodile).
has_scales(snake).
Rules:

1. An animal is a bird if it has feathers and lays eggs.
2. An animal is a reptile if it has scales and lays eggs.

Queries:

1. Is an eagle a bird?
2. Is a crocodile a reptile?

4.Write a Prolog program to classify plants based on their features.

Predicates:

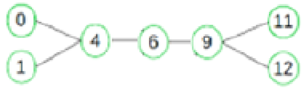
- classify(Plant, Type): Classifies the plant.
- feature(Plant, Property): Describes the properties of the plant.

Facts:
feature(rose, flowering).
feature(rose, thorny).
feature(cactus, succulent).
feature(cactus, thorny).
feature(mango, fruit_bearing).
feature(mango, woody).
Rules:
classify(rose, shrub) :- feature(rose, flowering), feature(rose, thorny).
classify(cactus, succulent) :- feature(cactus, succulent).
classify(mango, tree) :- feature(mango, fruit_bearing), feature(mango, woody).

Laboratory Exam Component - 100 marks			
Programs (4)			
Description	Marks Distribution	Total Marks	Marks Obtained
Aim & Algorithm	2.5*4 = 10 Marks	10	
Programs	10*4 = 30 Marks	40	
Debugging	2.5*4 = 10 Marks	10	
Input & Output	05*4 = 20 Marks	20	
Result	2.5*4 = 10 Marks	10	
GitHub content	10 Marks	10	
Total Lab Exam Marks		100	

Set 10

1. Write a python program to find goal state 6 using bidirectional search for the given graph.



2. Write a Python program to implement a two-layer feed-forward neural network to solve the XOR problem. The network should:
   Use a single hidden layer with two neurons.
   Use the sigmoid activation function.
   Train the network using the backpropagation algorithm.
3. Write a Prolog program to predict weather conditions based on observed indicators.
   Predicates:
   hypothesis(Location, Condition): Predicts the weather condition for a location.
   symptom(Location, Indicator): Observes weather indicators for a location.
   Facts:
   symptom(chennai, humid).
   symptom(chennai, cloudy).

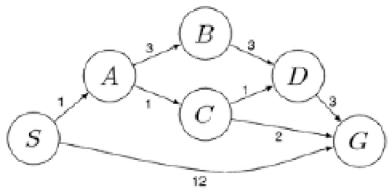
   symptom(delhi, sunny).
   symptom(delhi, dry).
   symptom(ooty, cold).
   symptom(ooty, foggy).
   Rules:
   hypothesis(chennai, rainy) :- symptom(chennai, humid), symptom(chennai, cloudy).
   hypothesis(delhi, clear) :- symptom(delhi, sunny), symptom(delhi, dry).
   hypothesis(ooty, foggy_weather) :- symptom(ooty, cold), symptom(ooty, foggy).

4. Write a Prolog program to predict a student's academic performance based on their behavior and attendance.
   Predicates:
   performance(Student, Level): Predicts the performance level of a student.
   attribute(Student, Trait): Describes the traits of a student.
   Facts:
   attribute(john, hardworking).
   attribute(john, regular).
   attribute(sarah, irregular).
   attribute(sarah, average).
   attribute(mike, hardworking).
   attribute(mike, irregular).
   Rules:
   performance(Student, excellent) :- attribute(Student, hardworking), attribute(Student, regular).
   performance(Student, good) :- attribute(Student, hardworking), attribute(Student, irregular).
   performance(Student, average) :- attribute(Student, average).

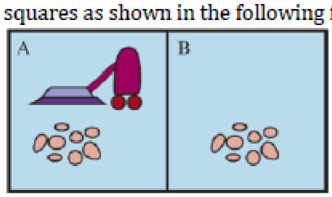
Laboratory Exam Component - 100 marks			
Programs (4)			
Description	Marks Distribution	Total Marks	Marks Obtained
Aim & Algorithm	2.5*4 = 10 Marks	10	
Programs	10*4 = 30 Marks	40	
Debugging	2.5*4 = 10 Marks	10	
Input & Output	05*4 = 20 Marks	20	
Result	2.5*4 = 10 Marks	10	
GitHub content	10 Marks	10	
Total Lab Exam Marks		100	

SET:11

1. Write a Python program for uniform cost search to solve the tree starting from S as initial node reach the goal node G



2. Write a Python program to build a Vacuum World with simple reflex agent for the A and B squares as shown in the following figure.



3. Write a Prolog program with the following facts, Dog-Name, Size-Small, Medium and Big dog (fido), dog(rover), dog(jane), dog(tom), dog(fred), dog(henry), dog(fido), cat(mary), cat(harry), cat(bill), cat(steve), small(henry), medium(harry), medium(fred), large(fido), large(mary), large(tom), large(fred), large(steve), large(jim), large(mike).

4. Write a Prolog program for a Planets Database with the following facts.
orbits(mercury, sun).
orbits(venus, sun).
orbits(earth, sun).
orbits(mars, sun).
orbits(moon, earth).
orbits(phobos, mars).
orbits(deimos, mars).

Laboratory Exam Component - 100 marks			
Programs (4)			
Description	**Marks Distribution**	**Total Marks**	**Marks Obtained**
Aim & Algorithm	2.5*4 = 10 Marks	10	
Programs	10*4 = 30 Marks	40	
Debugging	2.5*4 = 10 Marks	10	
Input & Output	05*4 = 20 Marks	20	
Result	2.5*4 = 10 Marks	10	
GitHub content	10 Marks	10	
Total Lab Exam Marks		**100**	

SET:12

1. Write a Python program, to calculate the optimal solution using Manhattan distance for the given 8 puzzle problem.

1	5	3
	4	2
7	8	6

Initial Sate

1	2	3
4	5	6
7	8	

Goal State

2. Write a Python program, to evaluate the 8-Queens problem using Genetic algorithms Propose a chromosome representation and crossover and mutation mechanisms tha generate result in exactly one queen per column and one queen per row.

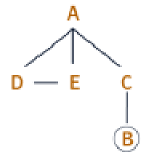
3. Write a Prolog program for the forward chaining using following facts
rainy(chennai).
rainy(coimbatore).
rainy(ooty).
cold(ooty).

4. Write a Prolog program to predict Fruits and its color using Backtracking
colour(cherry, red).
colour(banana, yellow).
colour(apple, red).
colour(apple, green).
colour(orange, orange).
colour(X, unknown).

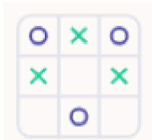
Laboratory Exam Component - 100 marks			
Programs (4)			
Description	**Marks Distribution**	**Total Marks**	**Marks Obtained**
Aim & Algorithm	2.5*4 = 10 Marks	10	
Programs	10*4 = 30 Marks	40	
Debugging	2.5*4 = 10 Marks	10	
Input & Output	05*4 = 20 Marks	20	
Result	2.5*4 = 10 Marks	10	
GitHub content	10 Marks	10	
Total Lab Exam Marks		**100**	

SET:13

1. Write a Python program to implement Depth First Search technique for the given state space graph.



2. Write a Python Program for Minimax search tree starting from this state, and ending in terminal nodes. Show the utility value for each terminal node. Also show which move the Minimax Algorithm decides to play. Utility values are +1 if X wins, 0 for a tie, and -1 if O wins. Assume that O makes the next move (O is the MAX player), use theTic-Tac-Toe board state as shown below

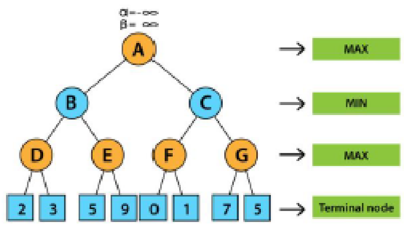


3. Write a Prolog program to implement pattern matching with the following facts.
first_name(tonyblair, tony).
first_name(georgebush, georgedubya).
second_name(tonyblair, blair).
second_name(georgebush, bush).
4. Write a Prolog program to implement Family Relation with the following facts.
female(sarah). female(rebekah). female(hagar_concubine).
female(milcah). female(bashemath). female(mahalath).
female(first_daughter). female(rachel). female(labans_wife).
male(terah). male(abraham). male(nahor).
male(haran). male(isaac). male(ismael).
male(uz). male(kemuel).

Laboratory Exam Component - 100 marks			
Programs (4)			
Description	**Marks Distribution**	**Total Marks**	**Marks Obtained**
Aim & Algorithm	2.5*4 = 10 Marks	10	
Programs	10*4 = 30 Marks	40	
Debugging	2.5*4 = 10 Marks	10	
Input & Output	05*4 = 20 Marks	20	
Result	2.5*4 = 10 Marks	10	
GitHub content	10 Marks	10	
Total Lab Exam Marks		**100**	

SET:14

1. Write a Python program for the following game tree using Alpha-Beta Pruning.



2. Design a web page of your own to show how Anchor tags and titles are optimized by Search Engine Optimization.

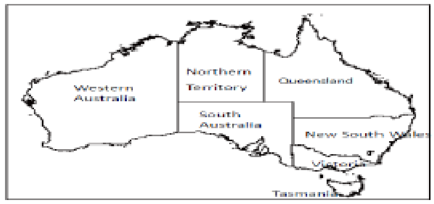
3. Write a Prolog program to find items and its location using following facts
   location(desk, office).
   location(apple, kitchen).
   location(flashlight, desk).

4. Write a Prolog program to predict the Causes of Disease with the following predicates and clauses.
   predicates
     hypothesis(name,disease)
     symptom(name,indication)
   clauses
     symptom(amit,fever).
     symptom(amit,rash).

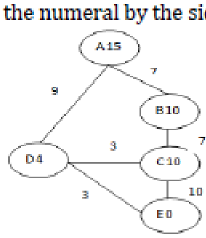
Laboratory Exam Component - 100 marks			
Programs (4)			
Description	**Marks Distribution**	**Total Marks**	**Marks Obtained**
Aim & Algorithm	2.5*4 = 10 Marks	10	
Programs	10*4 = 30 Marks	40	
Debugging	2.5*4 = 10 Marks	10	
Input & Output	05*4 = 20 Marks	20	
Result	2.5*4 = 10 Marks	10	
GitHub content	10 Marks	10	
Total Lab Exam Marks		**100**	

SET:15

1. Write a Python program to implement a Map Colouring problem with the following data. Given a map of countries, and a fixed set of colours {Red, Green, Blue}, assign a colour to each region in the map in such a way that no two adjacent regions have the same colour. Solve the above Map Colouring problem using backtracking in Constraint Satisfaction Procedure.



2. Write a Python program to apply A* search for the given tree, in which 'A' is the initial state and 'E' goal state. For each node in each stage, obtain cost estimates where g(n) is the numeral by the side of an arc and h(n) is the numeral at the node.



3. Write a Prolog program to implement Inference Engine with the following facts
   interpret(true) :- !.
   interpret((GoalA,GoalB)) :- !,
   interpret(GoalA),
   interpret(GoalB).

4. Write a Prolog program to find the number of vowels in the following fact.
   This is my first Degree in Saveetha School of Engineering.

Laboratory Exam Component - 100 marks			
Programs (4)			
Description	**Marks Distribution**	**Total Marks**	**Marks Obtained**
Aim & Algorithm	2.5*4 = 10 Marks	10	
Programs	10*4 = 30 Marks	40	
Debugging	2.5*4 = 10 Marks	10	
Input & Output	05*4 = 20 Marks	20	
Result	2.5*4 = 10 Marks	10	
GitHub content	10 Marks	10	
Total Lab Exam Marks		**100**	

Set-2

1. WATER JUG PROBLEM – PYTHON

Aim

To write a Python program to solve the Water Jug problem using Breadth First Search (BFS) and measure exactly 3 litres of water using a 5-litre jug and a 4-litre jug.

Procedure

1. Start with both jugs empty.
2. Represent the state as (x, y), where x is the amount of water in the 5-litre jug and y is the amount in the 4-litre jug.
3. Generate new states by filling, emptying, and transferring water between the jugs.
4. Use BFS to explore the possible states.

5. Avoid visiting the same state repeatedly.
6. Stop when either jug contains exactly 3 litres.
7. Display the sequence of states.

Algorithm

1. Initialize the starting state as (0,0).
2. Create a queue and add the initial state.
3. Remove a state from the queue.
4. Check whether either jug contains 3 litres.
5. If not, generate all possible operations:
   - Fill 5-litre jug.
   - Fill 4-litre jug.
   - Empty 5-litre jug.
   - Empty 4-litre jug.
   - Pour 5-litre jug into 4-litre jug.
   - Pour 4-litre jug into 5-litre jug.
6. Add unvisited states to the queue.
7. Repeat until the goal state is reached.
8. Print the solution path.

Code:

```
from collections import deque

def water_jug():
    queue = deque([((0, 0), [])])
    visited = set()

    while queue:
        (a, b), path = queue.popleft()

        if (a, b) in visited:
            continue

        visited.add((a, b))

        if a == 3 or b == 3:
            print("Solution:")
            for state in path + [(a, b)]:
                print(state)
            return

        states = [
            (5, b),
            (a, 4),
```

```
        (0, b),
        (a, 0)
    ]

    amount = min(a, 4 - b)
    states.append((a - amount, b + amount))

    amount = min(b, 5 - a)
    states.append((a + amount, b - amount))

    for state in states:
        if state not in visited:
            queue.append((state, path + [(a, b)]))

water_jug()
```

output:

Result:

Thus, the Water Jug problem was successfully solved using BFS, and exactly 3 litres of water was measured.

2. MISSIONARIES AND CANNIBALS – PYTHON

Aim

To write a Python program to solve the Missionaries and Cannibals problem using Breadth First Search, while satisfying all the safety conditions.

Procedure

1. Consider 3 missionaries and 3 cannibals on one side of the river.
2. The boat can carry a maximum of two people.
3. Represent each state using (M, C, B).
4. Generate all possible movements of one or two people.
5. Check whether the new state is safe.
6. Reject states where cannibals outnumber missionaries when missionaries are present.
7. Use BFS to find a valid path.
8. Continue until everyone reaches the opposite side.

Algorithm

1. Set the initial state as (3,3,1).
2. Set the goal state as (0,0,0).
3. Define possible boat movements:
   - One missionary.
   - Two missionaries.
   - One cannibal.
   - Two cannibals.
   - One missionary and one cannibal.
4. Check whether each new state is valid.
5. Store visited states to avoid repetition.
6. Use a queue to perform BFS.
7. Continue until the goal state is reached.
8. Display the complete solution path.

Code:

```
from collections import deque

def safe(m, c):
    if m > 0 and c > m:
        return False

    rm = 3 - m
    rc = 3 - c

    if rm > 0 and rc > rm:
        return False

    return True


def solve():
    start = (3, 3, 1)
    goal = (0, 0, 0)

    moves = [
        (1, 0),
        (2, 0),
        (0, 1),
        (0, 2),
        (1, 1)
    ]
```

```
    queue = deque([(start, [])])
    visited = set()

    while queue:
        state, path = queue.popleft()

        if state in visited:
            continue

        visited.add(state)

        if state == goal:
            for s in path + [state]:
                print(s)
            return

        m, c, boat = state

        for dm, dc in moves:
            if boat == 1:
                nm = m - dm
                nc = c - dc
                nb = 0
            else:
                nm = m + dm
                nc = c + dc
                nb = 1

            if 0 <= nm <= 3 and 0 <= nc <= 3:
                if safe(nm, nc):
                    new_state = (nm, nc, nb)

                    if new_state not in visited:
                        queue.append(
                            (new_state, path + [state])
                        )

solve()
```
output:

```
================================================================ RESTART: C:/Users/
(3, 3, 1)
(3, 1, 0)
(3, 2, 1)
(3, 0, 0)
(3, 1, 1)
(1, 1, 0)
(2, 2, 1)
(0, 2, 0)
(0, 3, 1)
(0, 1, 0)
(1, 1, 1)
(0, 0, 0)
>>> |
```

Result:

Thus, all 3 missionaries and 3 cannibals successfully crossed the river without violating any of the given conditions.

3. MORTAL OR NOT – PROLOG

Aim

To write a Prolog program to determine whether a given person is mortal or not using facts and logical rules.

Procedure

1. Define the persons Socrates, Einstein and Alexander as men.
2. Define a rule stating that every man is mortal.
3. Give a query to Prolog for a particular person.
4. Prolog checks the facts and applies the rule.
5. If the person is a man, Prolog returns true.
6. If the person is not a man, Prolog returns false.

Algorithm

1. Define man(socrates).
2. Define man(einstein).
3. Define man(alexander).
4. Define the rule mortal(X) :- man(X).
5. Query whether the required person is mortal.
6. Display the result.

Code:

```
man(socrates).
man(einstein).
man(alexander).

mortal(X) :-
  man(X).
```

Output:

Result:

Thus, the Prolog program successfully determines that Socrates, Einstein and Alexander are mortal based on the given facts and rule.

4. COUNT NUMBER OF VOWELS – PROLOG

Aim

To write a Prolog program to count the total number of vowels present in the given sentence:

"I am an brilliant student in Saveetha University"

Procedure

1. Define the five vowels a, e, i, o, u.
2. Read the given sentence.
3. Convert the sentence into lowercase.
4. Convert the sentence into a list of characters.
5. Check each character one by one.
6. If the character is a vowel, increase the count.
7. If it is not a vowel, continue without increasing the count.
8. Repeat until all characters are checked.
9. Display the total number of vowels.

Algorithm

1. Define vowel(a), vowel(e), vowel(i), vowel(o) and vowel(u).
2. Define the base case for an empty list as zero.
3. Check the first character of the list.
4. If it is a vowel, add 1 to the count.
5. If it is not a vowel, keep the count unchanged.
6. Recursively process the remaining characters.
7. Convert the input sentence into lowercase characters.
8. Display the final vowel count

Code:

vowel(a).

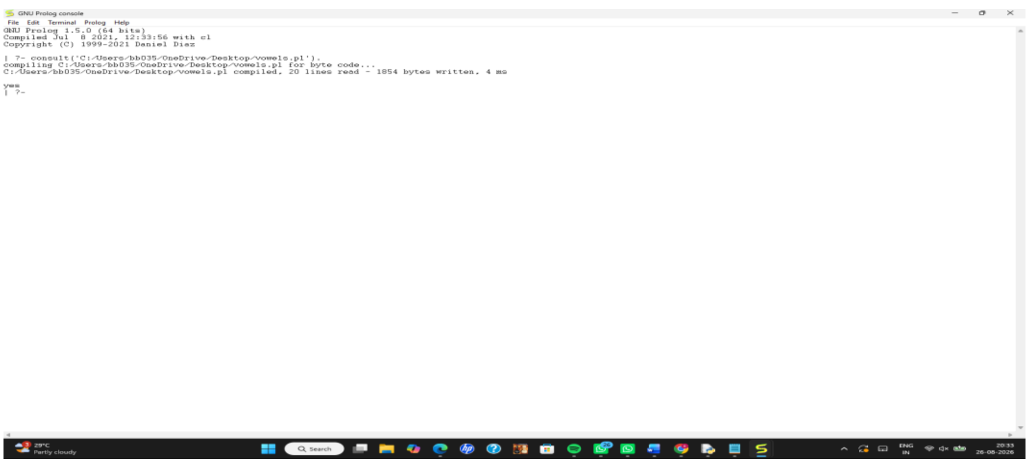
vowel(e).

vowel(i).

vowel(o).

```
vowel(u).

count_vowels([], 0).

count_vowels([H|T], N) :-
    vowel(H),
    count_vowels(T, N1),
    N is N1 + 1.

count_vowels([H|T], N) :-
    \+ vowel(H),
    count_vowels(T, N).

count_sentence(S, N) :-
    string_lower(S, L),
    string_chars(L, C),
    count_vowels(C, N)..
```

output:



Result:

The program successfully counts the vowels in the given sentence, and the total number of vowels is 16.